

Beerkanwar Singh

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EDUCATION

Dr. B. R. Ambedkar National Institute of Technology

Bachelor of Technology in Computer Science Engineering

CGPA: 7.61/10.0

Jalandhar, India

2023 – 2027 (Expected)

GAME DESIGN & TECHNICAL SKILLS

- **Game Design & Prototyping:** Level Design, Mechanic Design & Iteration, Difficulty/Progression Balancing, Playtesting, Game Feel & Player Engagement, State-Based Gameplay Systems
- **Engines & Tools:** Unity (URP, ML-Agents, Netcode for GameObjects), Phaser, Tiled (level design), Git, .NET 8, WPF
- **AI-Assisted Workflow:** GitHub Copilot, Google Gemini, LLM tools for design brainstorming & code iteration; AI image generation for game assets
- **Programming Languages:** C++, C#, JavaScript, Python
- **CS Fundamentals:** Data Structures & Algorithms, OOP, Design Patterns, Event-Driven Architecture, Finite State Machines, MVVM

EXPERIENCE

Aicade (Braincade Technologies Pvt. Ltd.)

Remote

Game Development Intern

Jun 2025 – Aug 2025

- Independently developed 21 HTML5 games (JavaScript, Phaser, HTML5, CSS3) based on reference games, concepts, and gameplay ideas provided by the team.
- Used AI-assisted development workflows (GitHub Copilot, Google Gemini) for code generation, iteration, debugging, and selected asset-generation tasks, while retaining ownership of final implementation.
- Iterated on gameplay, mechanics, and UI to turn reference concepts into functional, playable web games; completed and published all 21 games to Aicade's platform, contributing them to the company's RAG pipeline for its AI-driven game-development workflow.
- Received positive evaluations for the work produced; one shipped title remained among the top 5 games on the platform for approximately eight months, including after the internship ended.

PROJECTS

Snake — *Unity, C#, Mobile (Android – Google Play Store) & PC*

- Shipped a Snake game to the Google Play Store built on a `LinkedList<SnakeSegmentData> + HashSet<Vector2Int>` combination chosen for $O(1)$ head/tail manipulation and $O(1)$ self-collision detection.
- Implemented cross-platform input handling unifying keyboard input and swipe-gesture detection into a single buffered-input system that rejects illegal 180° reversals.
- Designed a score-driven dynamic difficulty formula, adaptive camera scaling for any screen size, and a self-contained autonomous "attract mode" AI snake with its own death/respawn lifecycle.

Kitchen Chaos — *Unity, C#, ScriptableObject*

- Designed a polymorphic counter/interaction system where 8 station types independently override shared `Interact/Interacted` contracts, eliminating the need for a centralized type-switch.
- Implemented independent per-object cooking state machines (`Idle`→`Frying`→`Fried`→`Burned`) driven by `ScriptableObject`-authored timers, and a fully data-driven recipe/ingredient system.
- Used `C#` events (including static, cross-instance events) to decouple UI, audio, and gameplay logic, and wrote a dedicated static-state reset manager to correctly handle Unity's cross-scene static delegate persistence bug.

Reinforcement Learning on Flappy Bird — *Unity ML-Agents (PPO), C#*

- Designed and trained a PPO reinforcement-learning agent to play Flappy Bird from a 4-dimensional continuous observation space and a 2-action discrete action space, training to convergence over 500,000 steps.
- Implemented event-driven reward shaping (sparse reward on obstacle-clear plus a small continuous survival bonus) reusing an existing gameplay event rather than adding separate instrumentation.
- Built a dual-mode agent script supporting both human keyboard control and trained-policy control from the same codebase for direct human-vs-policy comparison.

Bomb Massacre — *GMTK Game Jam – Unity, C#*

- Co-designed and built a "cook-to-expand" bomb mechanic with destructible tilemap terrain as the core gameplay hook, plus a single-scene level-select and win/lose game-loop manager, all under a strict jam deadline.

ACHIEVEMENTS

- **Problem-Solving Foundation:** Solved 250+ data structures and algorithms problems, supporting fast, structured creative problem-solving.
- **Winner, Team CTF Cybersecurity Competition** – cross-team collaboration under time pressure.